

OOSTENDE-BRUGES INTL, Belgium

WMO: 064070

Lat: 51.2003N

Lon: 2.8872E

Elev: 4

StdP: 101.28

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.1	-3.3	-8.5	1.8	-4.1	-6.3	2.2	-2.4	16.4	8.4	14.8	8.5	3.4	60	0.626

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.7	26.2	19.1	23.9	18.5	22.2	17.7	20.2	24.7	19.2	22.9	18.3	21.5	4.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.6	13.4	22.5	17.6	12.6	21.4	16.8	12.0	20.5	58.1	24.9	54.7	23.0	51.7	21.6	24.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.9	11.3	10.0	-7.8	31.9	2.9	1.9	-9.9	33.2	-11.6	34.3	-13.2	35.3	-15.3	36.7		
			-8.3	22.6	2.9	0.9	-10.4	23.3	-12.0	23.8	-13.6	24.4	-15.7	25.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.5	4.1	4.5	6.5	9.1	12.3	15.1	17.2	17.4	15.1	11.7	7.8	4.8
	DBStd	5.63	3.80	3.45	2.97	3.02	2.94	2.60	2.45	2.31	2.31	3.02	3.06	3.70
	HDD10.0	769	183	154	112	53	9	0	0	0	0	17	78	162
	HDD18.3	2902	440	387	366	278	188	103	53	47	101	205	316	418
	CDD10.0	953	1	1	4	24	81	153	222	228	153	71	12	2
	CDD18.3	46	0	0	0	0	2	7	17	17	3	0	0	0
	CDH23.3	266	0	0	0	1	15	47	91	93	18	1	0	0
	CDH26.7	63	0	0	0	0	2	12	21	25	3	0	0	0
Wind	WSAvg	4.8	5.6	5.4	5.3	4.7	4.7	4.5	4.3	4.2	4.4	4.7	4.9	5.3
Precipitation	PrecAvg	777	62	55	45	43	56	66	70	74	66	75	87	77
	PrecMax	962	120	97	93	78	143	148	160	158	168	150	174	182
	PrecMin	570	3	14	9	5	12	24	15	7	5	19	10	14
	PrecStd	106	28	25	24	21	28	31	34	45	43	32	36	42
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.7	13.3	17.1	22.0	25.4	28.7	29.4	30.0	25.8	21.7	16.1	13.4
		MCWB	10.9	9.9	11.6	14.2	16.9	19.6	20.3	21.0	19.0	17.0	14.1	11.7
	2%	DB	11.4	11.6	13.9	18.5	21.5	23.9	25.7	25.7	22.3	18.7	14.4	12.2
		MCWB	9.9	9.4	10.5	12.8	15.4	18.3	19.5	19.2	17.5	15.7	12.8	10.8
	5%	DB	10.1	10.4	12.1	15.8	18.9	21.3	23.4	23.2	20.6	17.1	13.3	11.2
		MCWB	8.8	8.6	9.5	11.7	14.5	17.0	18.5	18.8	16.8	14.8	11.8	9.8
	10%	DB	9.1	9.3	10.8	13.7	17.0	19.5	21.6	21.7	19.2	16.0	12.2	9.9
		MCWB	7.8	7.7	8.8	10.8	13.6	15.9	17.7	17.8	15.9	14.0	10.6	8.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.2	10.8	12.5	14.8	18.6	20.7	21.6	21.9	19.7	17.5	14.4	12.1
		MCDB	12.3	12.6	15.7	20.0	23.1	26.7	27.1	27.5	23.9	20.6	15.5	13.1
	2%	WB	10.1	9.8	11.0	13.3	16.1	18.7	20.1	20.3	18.4	16.2	13.0	11.1
		MCDB	11.3	11.2	13.4	17.5	20.2	23.1	24.7	24.3	21.5	18.2	14.3	12.2
	5%	WB	9.0	8.8	9.9	12.1	14.9	17.3	19.0	19.3	17.3	15.2	12.0	10.0
		MCDB	10.0	10.2	11.8	15.2	18.3	20.6	22.5	22.6	19.9	16.9	13.2	11.1
	10%	WB	7.9	7.9	9.0	11.0	13.9	16.3	18.1	18.2	16.4	14.3	10.8	8.7
		MCDB	9.0	9.2	10.6	13.4	16.8	19.2	21.3	21.1	18.8	15.9	12.0	9.8
Mean Daily Temperature Range	5% DB	MDBR	4.6	5.3	6.1	7.2	7.0	7.1	7.3	7.7	7.3	6.5	5.1	4.6
		MCDBR	5.0	6.5	8.6	10.9	10.8	11.0	10.8	10.9	10.1	7.5	5.6	4.7
		MCWBR	4.6	5.0	5.9	6.6	6.2	6.5	5.9	6.1	6.0	5.3	4.7	4.5
	5% WB	MCDBR	4.8	5.6	7.6	9.5	9.5	10.0	9.3	9.5	8.6	7.0	5.4	4.8
		MCWBR	4.6	4.7	5.5	6.0	5.8	6.2	5.6	5.7	5.5	5.3	4.8	4.6
Clear-Sky Solar Irradiance	taub	0.334	0.353	0.391	0.414	0.424	0.421	0.422	0.414	0.392	0.371	0.346	0.329	
	taud	2.368	2.352	2.251	2.220	2.245	2.274	2.279	2.320	2.368	2.402	2.379	2.346	
	Ebn at Noon	674	764	797	823	832	837	828	816	792	734	651	612	
	Edh at Noon	64	84	111	128	130	128	125	114	98	79	63	56	
All-Sky Solar Radiation	RadAvg	0.77	1.47	2.70	4.40	5.14	5.58	5.27	4.50	3.33	1.90	0.90	0.59	
	RadStd	0.07	0.27	0.34	0.41	0.38	0.45	0.53	0.38	0.22	0.16	0.09	0.07	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (70 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page